

Isolation and Characterization of Rhizospheric Bacteria from Heavy Metal Contaminated Soil of Savar, Dhaka and Evaluation of Their Potential Plant Growth Promoting Attributes

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ABSTRACT

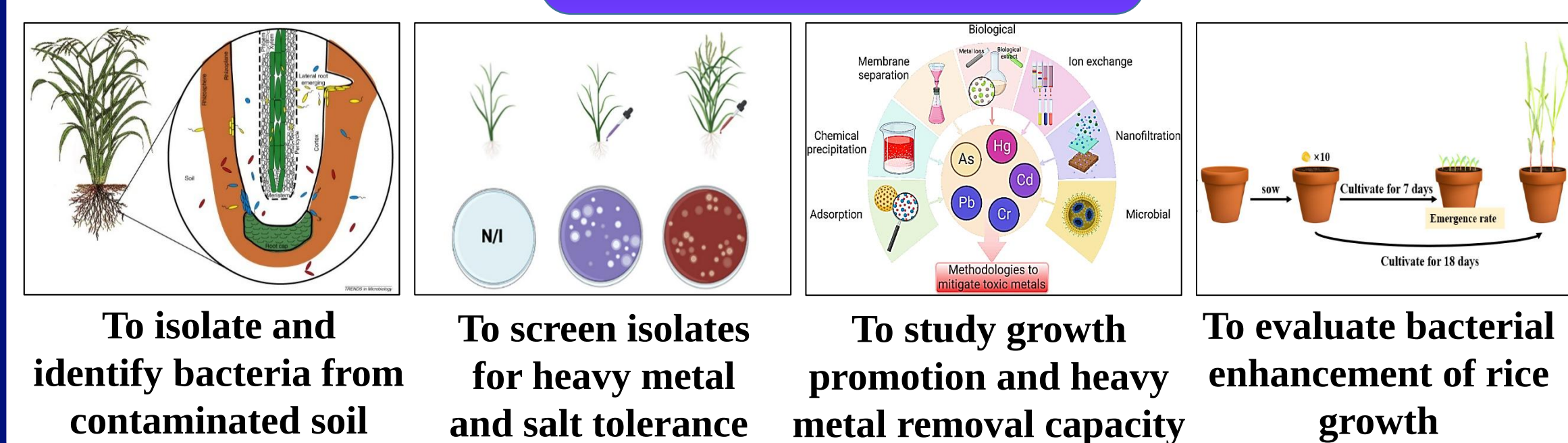
Soil pollution caused by heavy metals is a major problem in Bangladesh, especially in industrial zones like Savar and Dhaka. The main goal of this study is to isolate and characterize heavy metal-resistant PGPRs from the rhizosphere of vegetable plants from contaminated soil in Savar. In Hemayetpur, Savar, and Dhaka, various vegetable plants were irrigated by wastewater from the tannery industry, and rhizospheric bacteria were isolated and purified from the root-associated soil. Nine bacteria have been isolated and screened for plant growth-promoting (PGP) traits. These bacteria include nitrogen fixation, phosphate and zinc solubilization, ammonia production, indole-3-acetic acid (IAA) synthesis, and siderophore production. All of these bacteria were evaluated for their ability to survive against heavy metals such as cadmium (Cd), lead (Pb), and manganese (Mn) and their performance under salt stress. The heavy metal removal efficiency tests showed that Iso-6 was able to remove 63%, 84%, and 69% against Cd, Mn, and Pb, respectively. Iso-8 was also able to remove 7%, 86%, and 78% of Cd, Mn, and Pb, respectively. Controlled pot experiments on rice have shown that inoculation with these bacterial strains meaningfully increased plant growth compared to uninoculated controls. In contaminated soils, the potential of native heavy metal-resistant PGPR strains is highlighted in this study to enhance plant growth and productivity. Reducing the toxicity of heavy metals and enhancing plant growth—these two capabilities indicate that they have potential for agriculture and ecological restoration in polluted environments.

INTRODUCTION

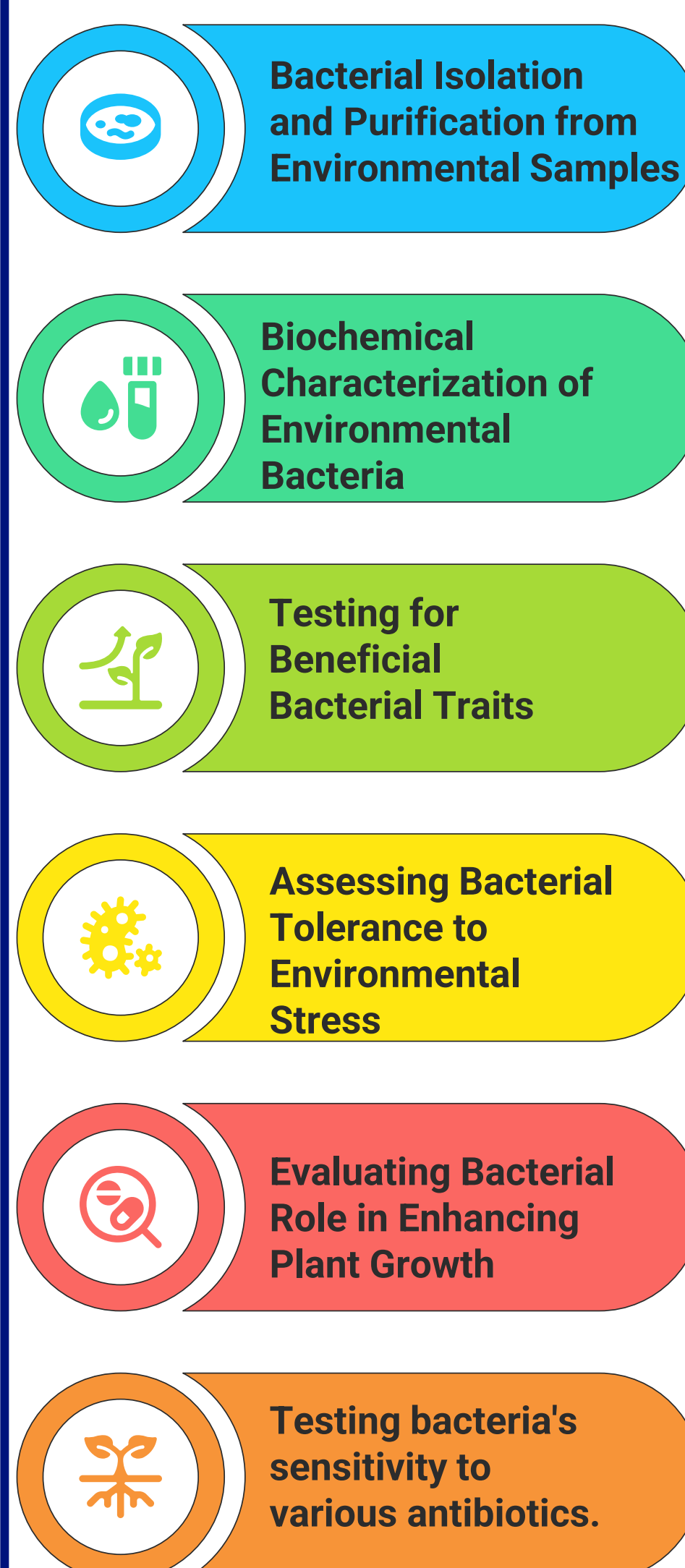
Rhizospheric bacteria, inhabiting the root-soil interface, play a crucial role in plant growth and health. These bacteria promote plant growth by fixing atmospheric nitrogen, solubilizing minerals, and producing phytohormones. Even Some rhizospheric bacteria contribute to phytoremediation. Additionally, they contribute to heavy metal detoxification by producing siderophores (1), which chelate toxic metals such as Cr, Cd, Pb, Mn, etc. reducing their bioavailability. These bacteria can reduce the availability of toxic metals through mechanisms like biosorption and bioaccumulation, helping to mitigate the harmful effects of heavy metals on plant growth.

By stabilizing soil nutrient balances and protecting plants from metal-induced salinity and nutrient deficiencies, rhizospheric bacteria support healthier plant growth even in contaminated soils. Their ability to detoxify heavy metals (2) also makes them valuable in bioremediation efforts, contributing to safer and more sustainable agricultural practices. They also act as biocontrol agents, suppressing harmful pathogens like *Fusarium oxysporum*, which causes Fusarium wilt in key crops (3).

OBJECTIVES

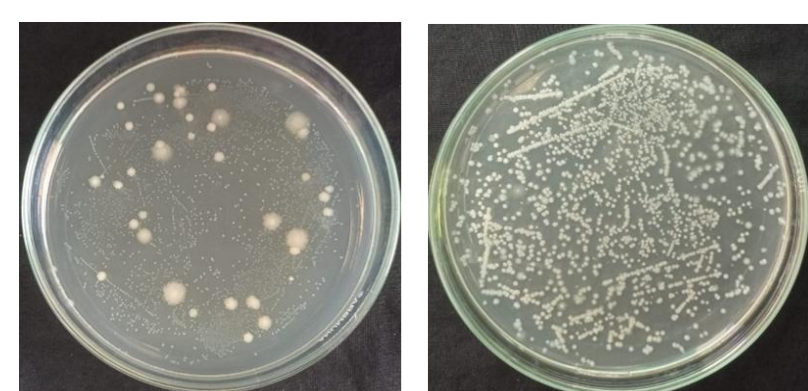


METHOD

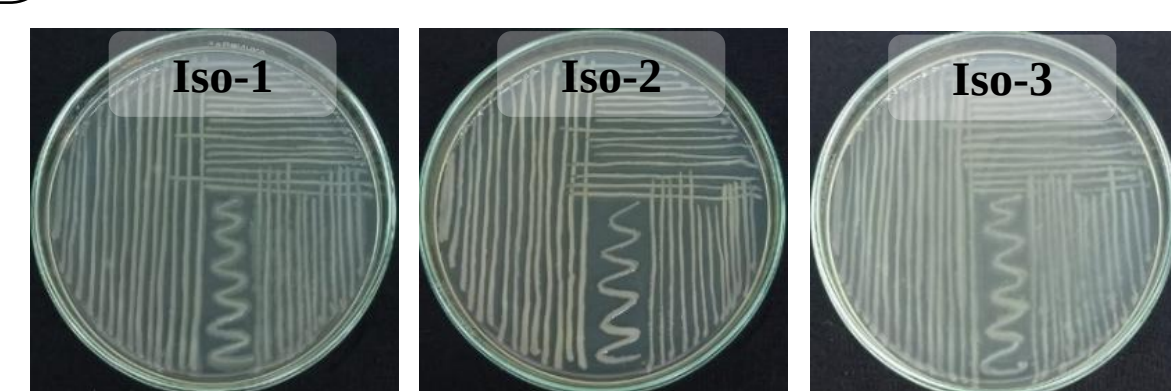


RESULTS and DISCUSSION

Isolation of Bacteria

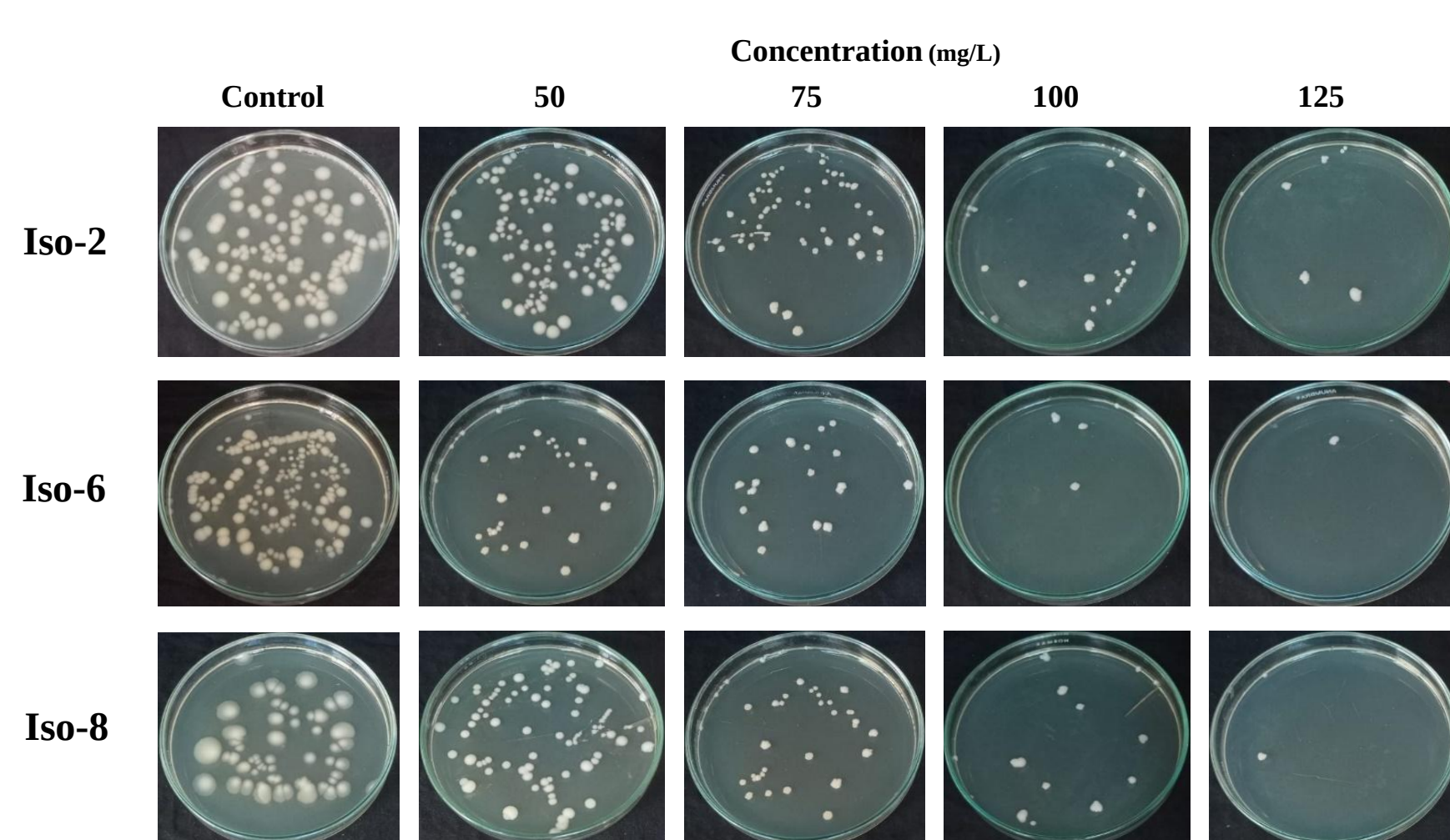


Spreading Plates

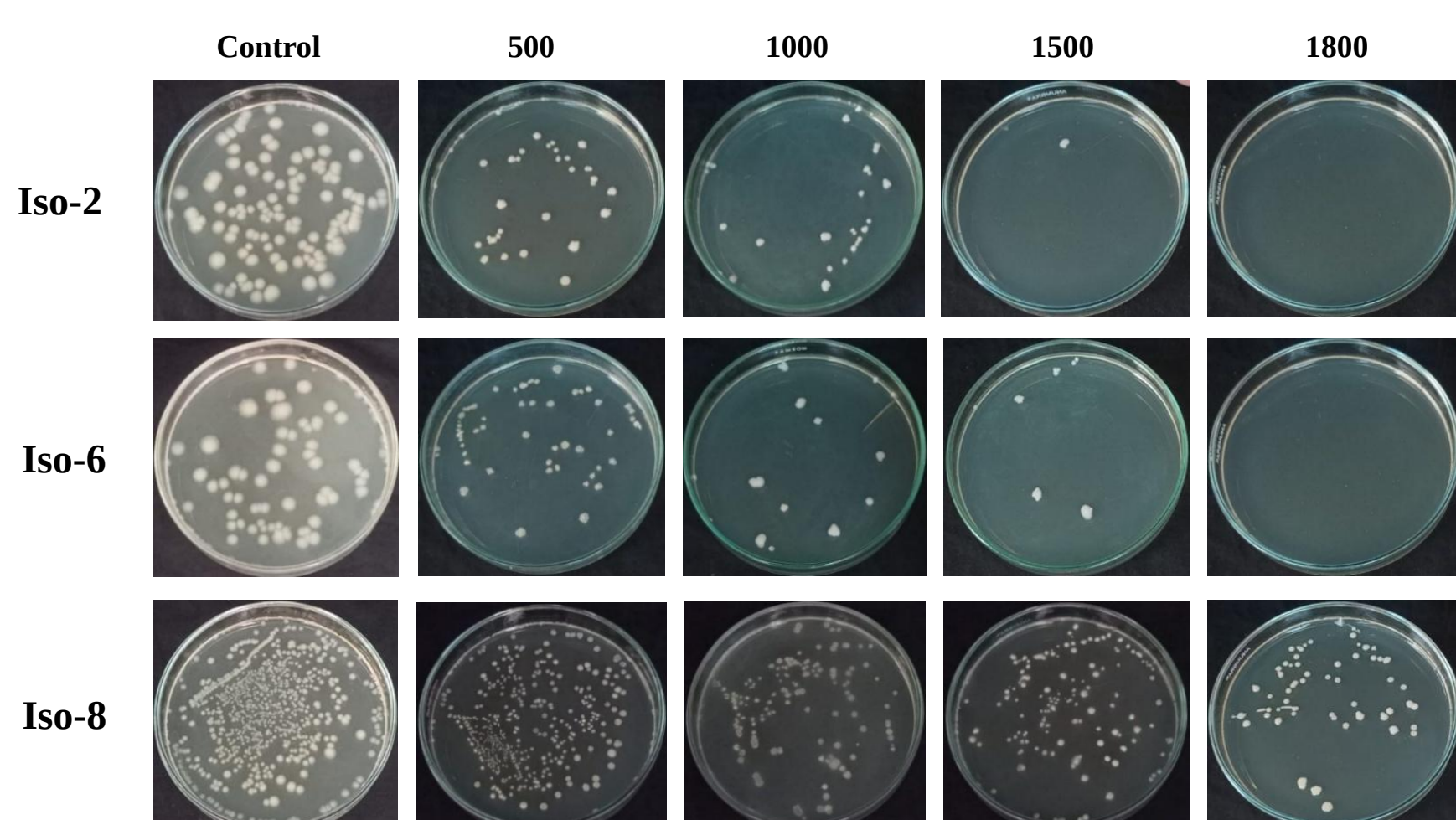


Streaking Plates

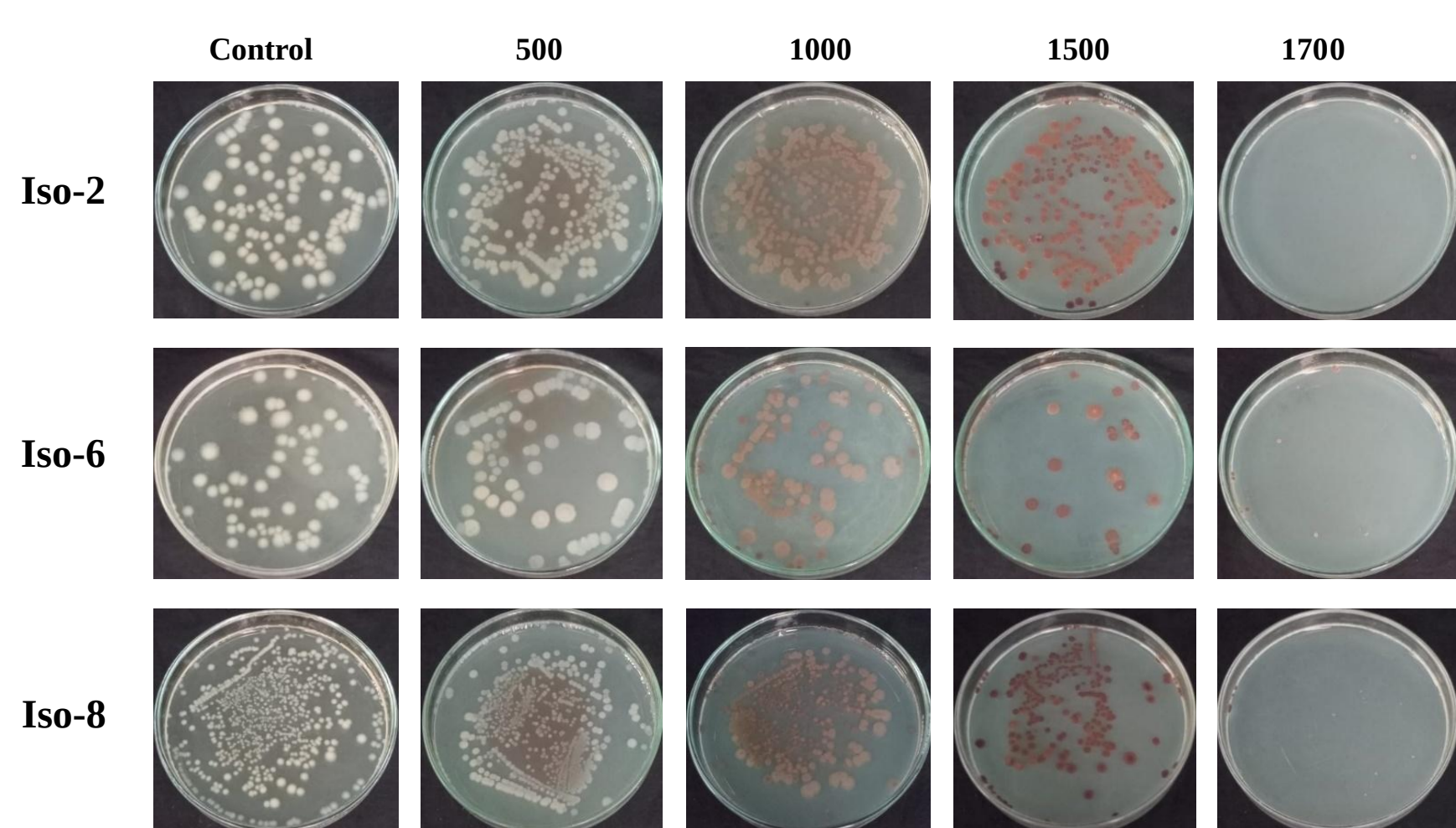
Heavy Metals Resistant Assay of Isolated Bacteria



Assay of Growth Rate of Isolates in The Presence of Cadmium (mg/L)

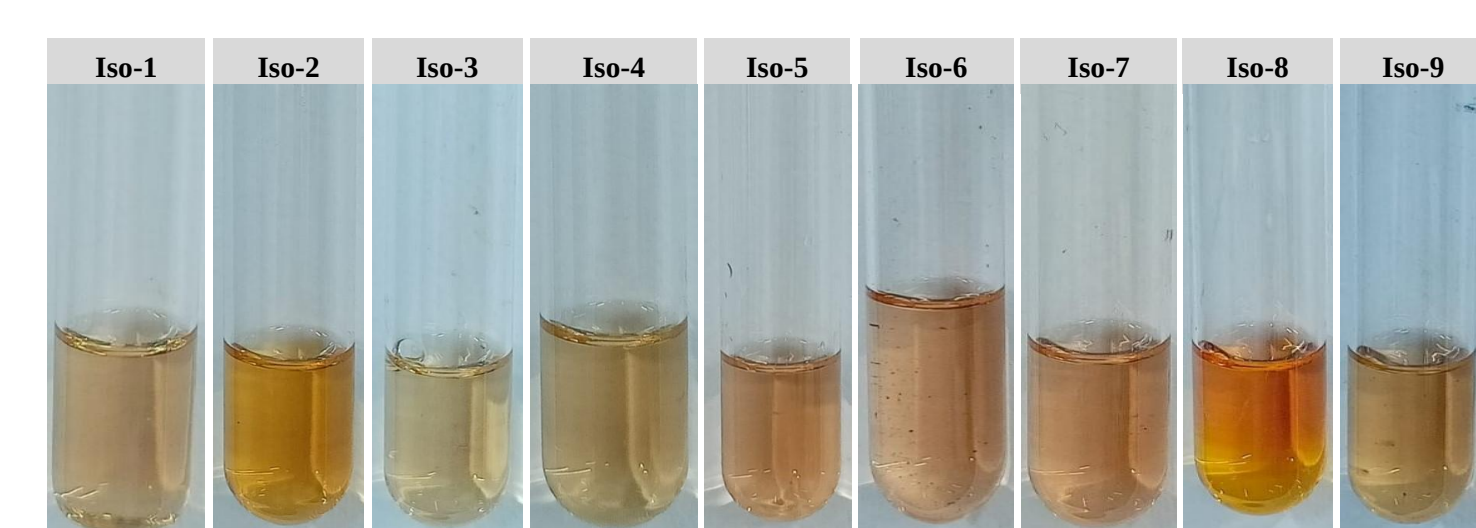


Assay of Growth Rate of Isolates in The Presence of Manganese (mg/L)

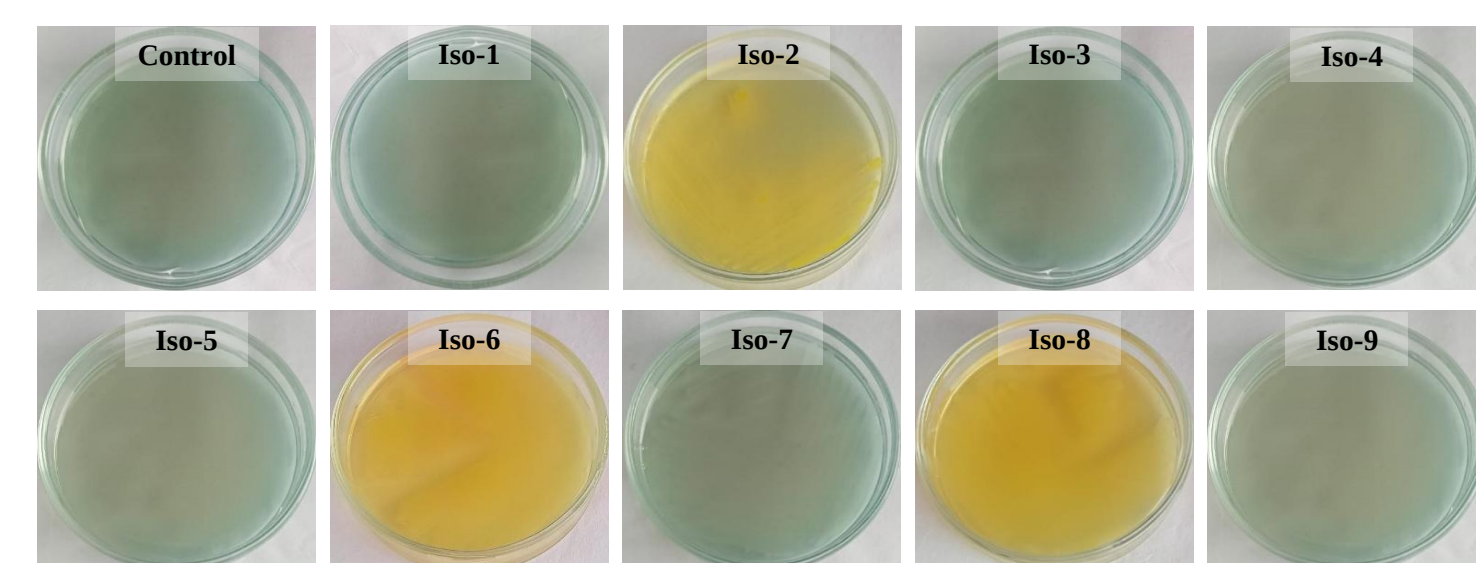


Assay of Growth Rate of Isolates in The Presence of Lead (mg/L)

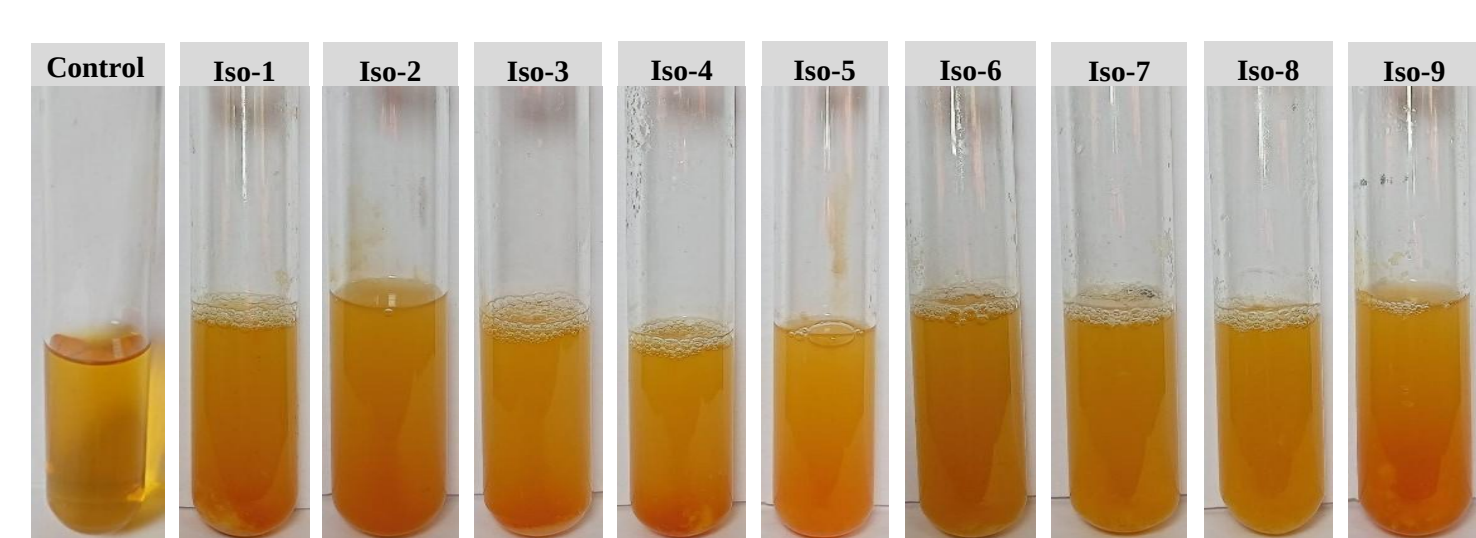
Plant Growth Promoting Traits of Isolated Bacteria



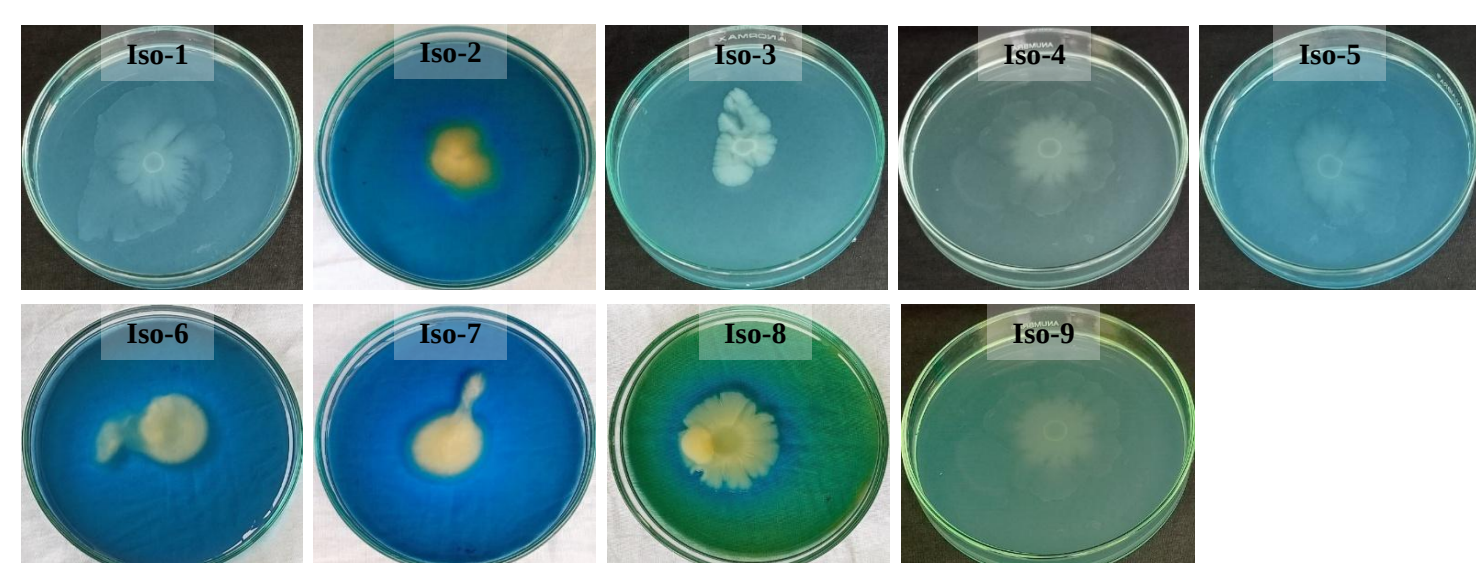
Indole-3-Acetic Acid (IAA) Assay



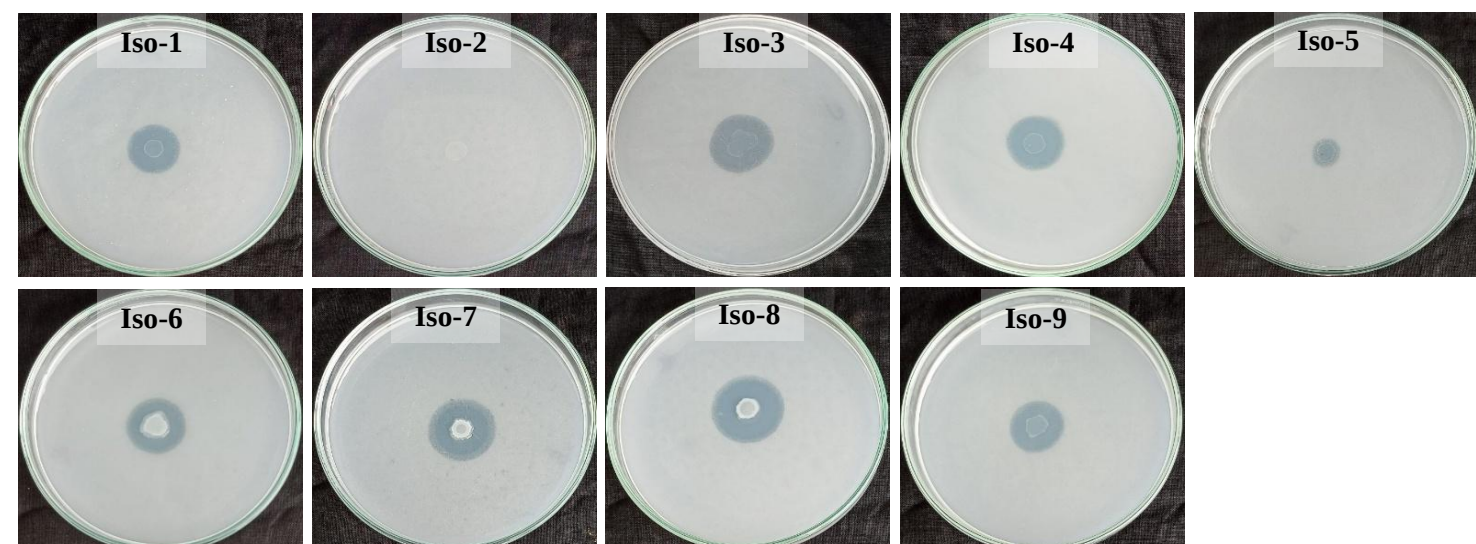
Nitrogen Fixation Test



Ammonia Test

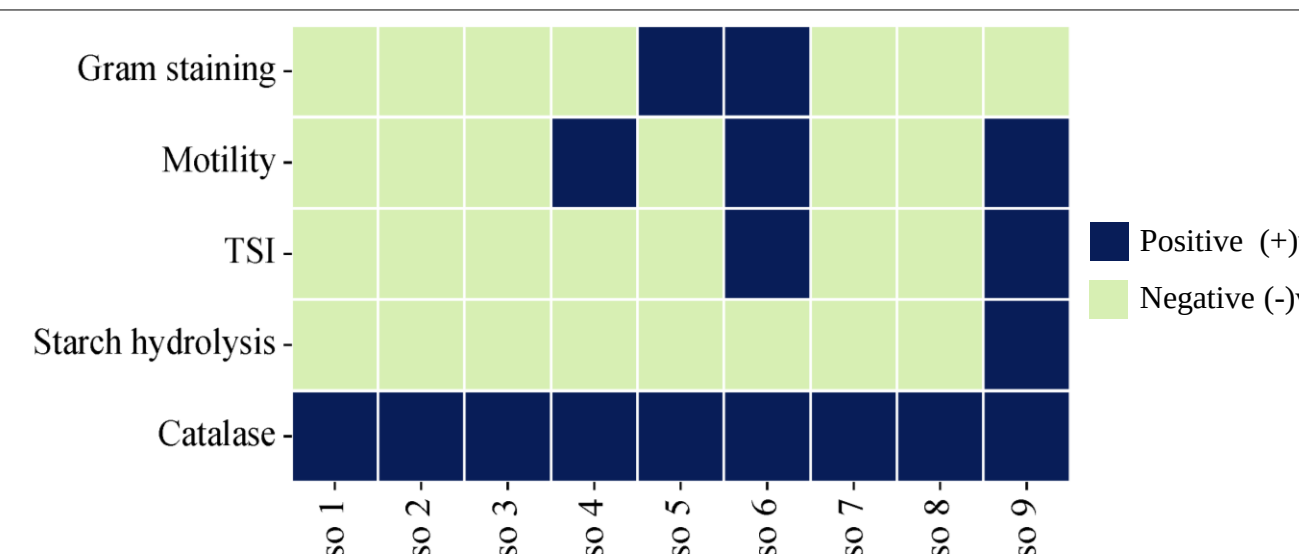


Siderophore Assay

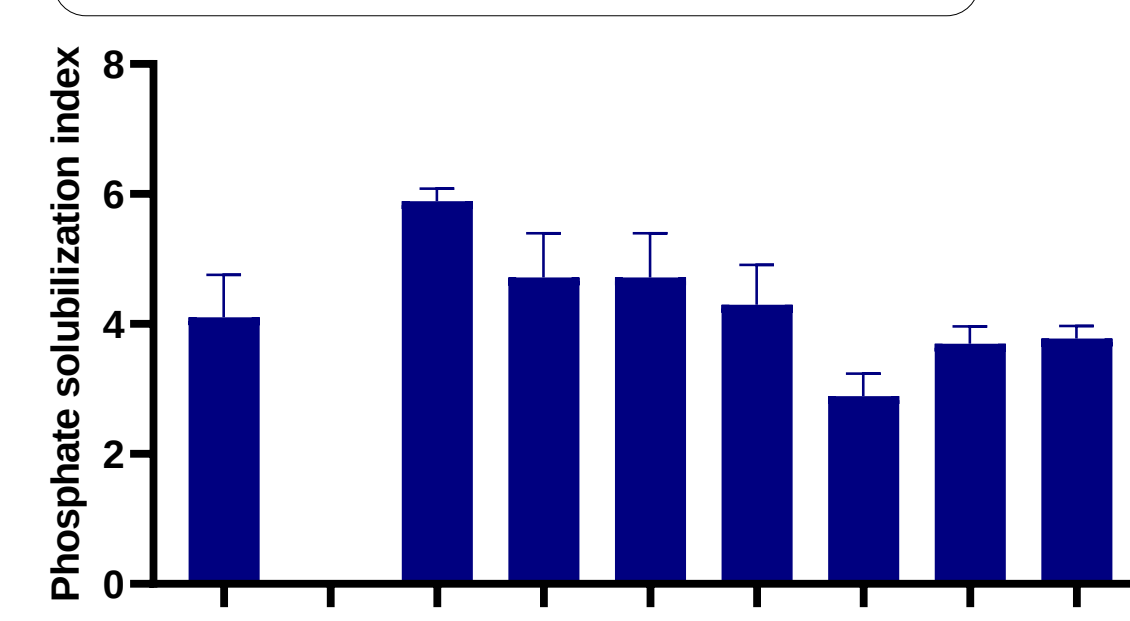


Zinc Solubilizing Test

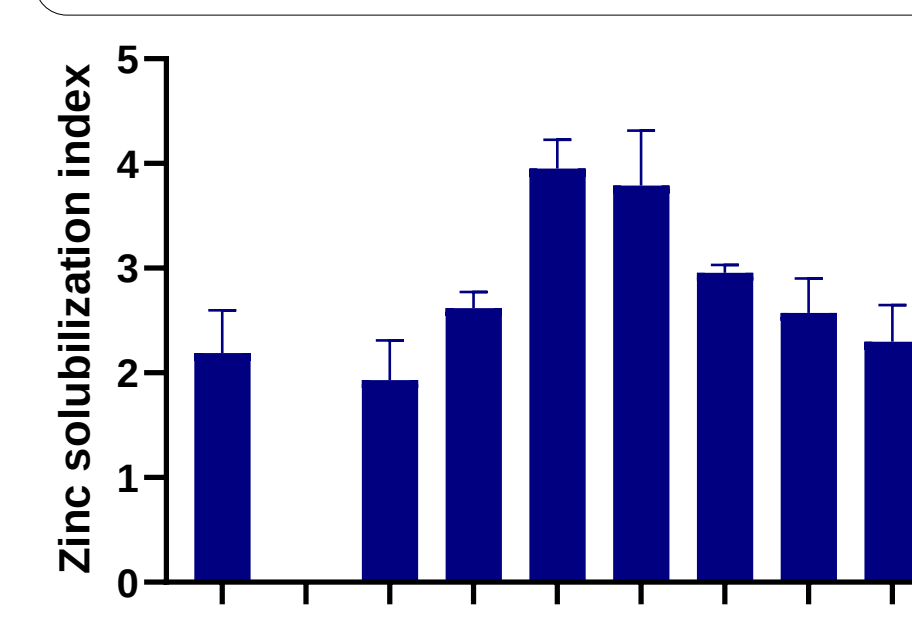
Summary of The Results of Morphological and Biochemical Test



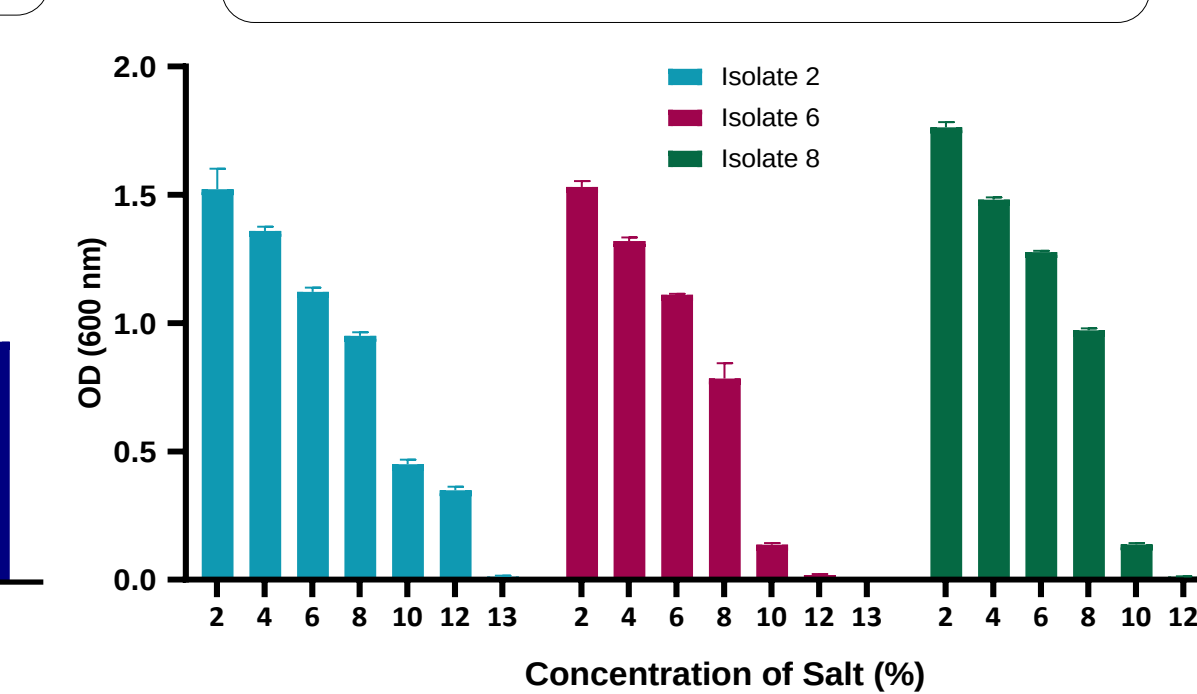
Graphical representation of Phosphate Solubilization Assay



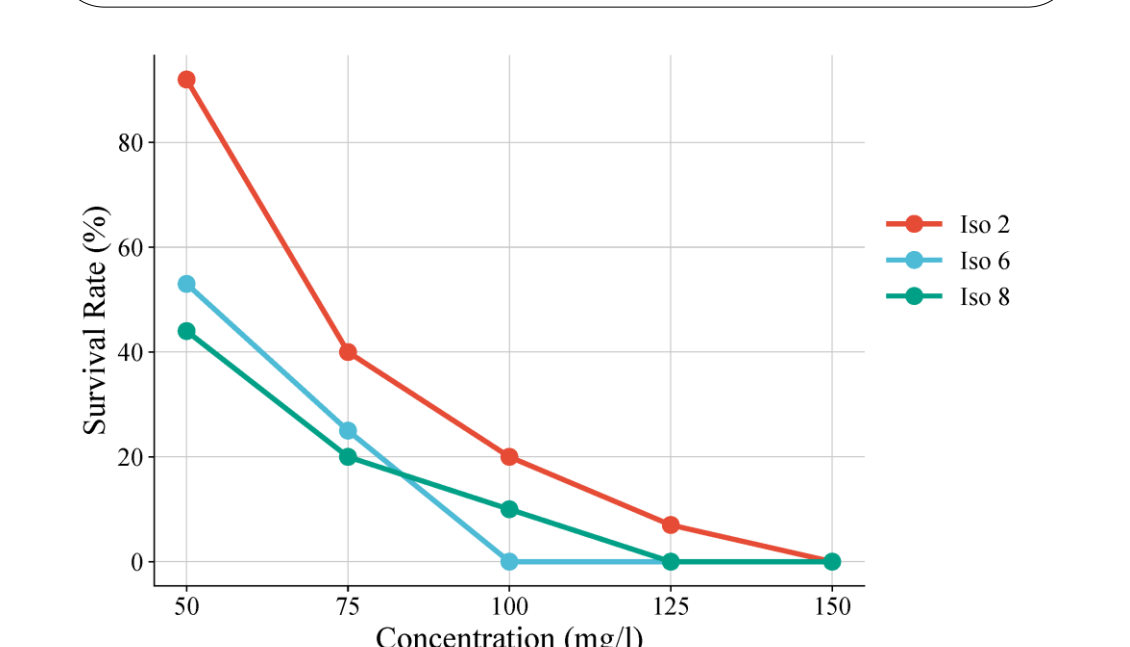
Graphical Representation of Zinc Solubilization Assay



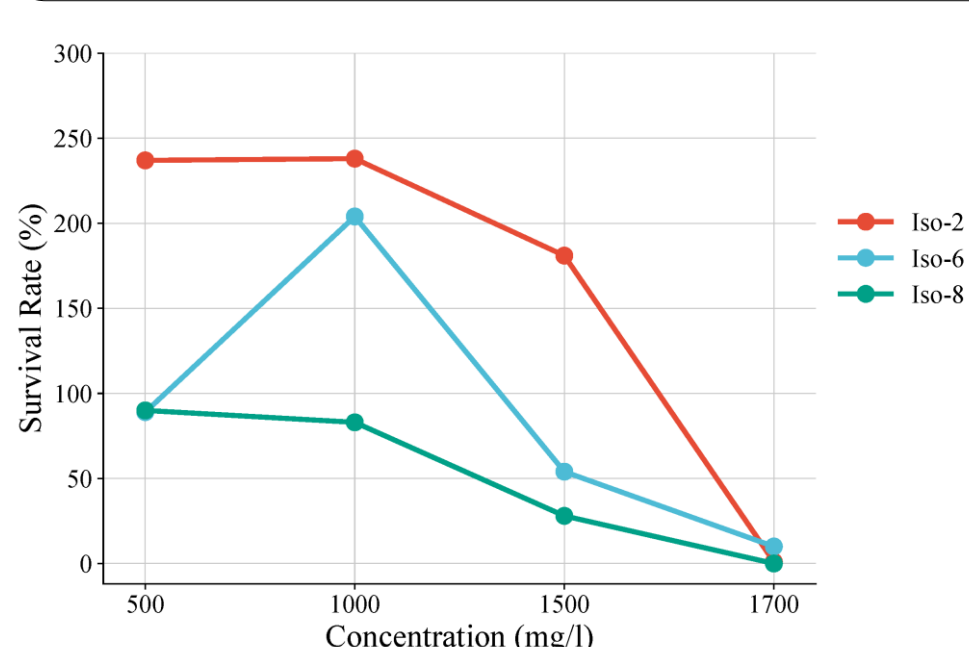
Graphical representation of Salt Tolerance Test



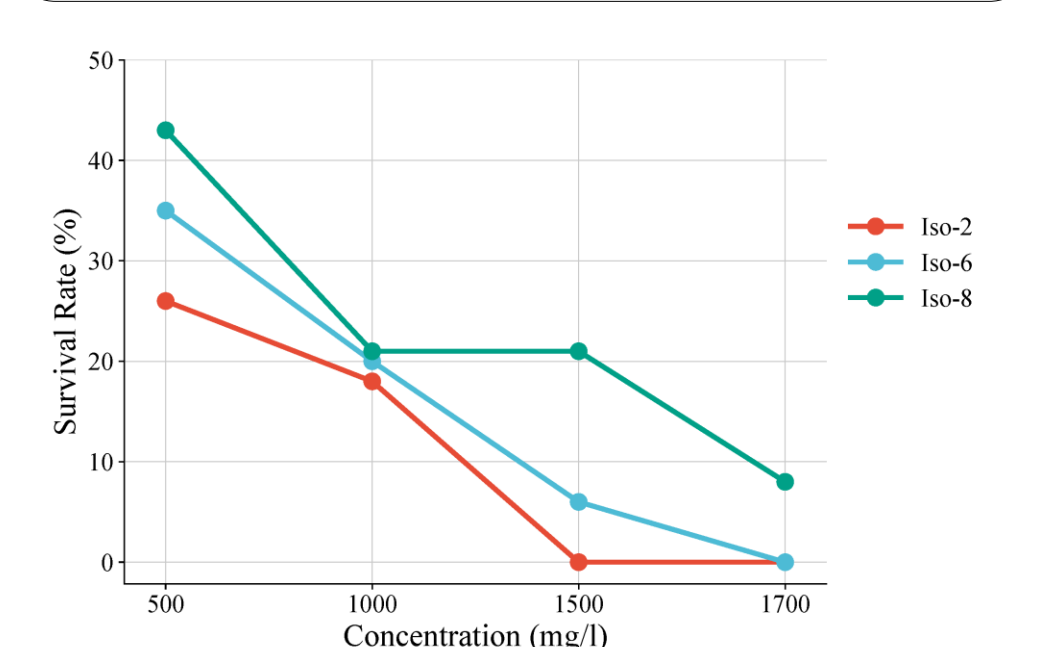
Graphical Representation of Survival Rate (%) of Isolated Bacteria in Presence of Cd (mg/L)



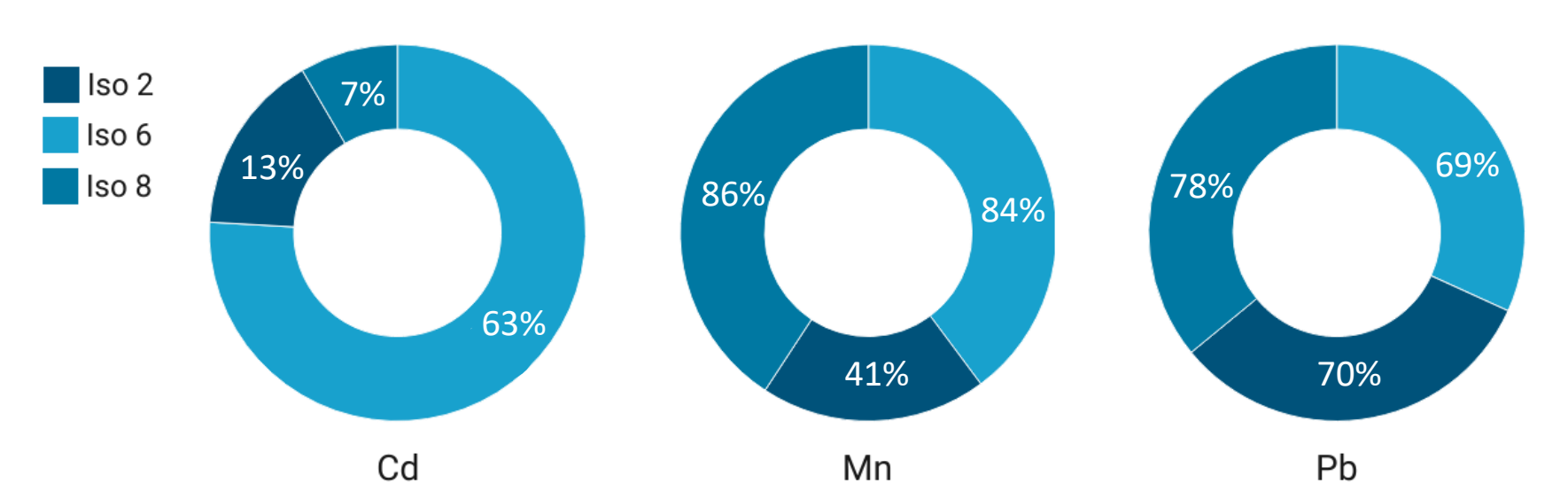
Graphical representation of Survival Rate (%) of Isolated Bacteria in Presence of Pb (mg/L)



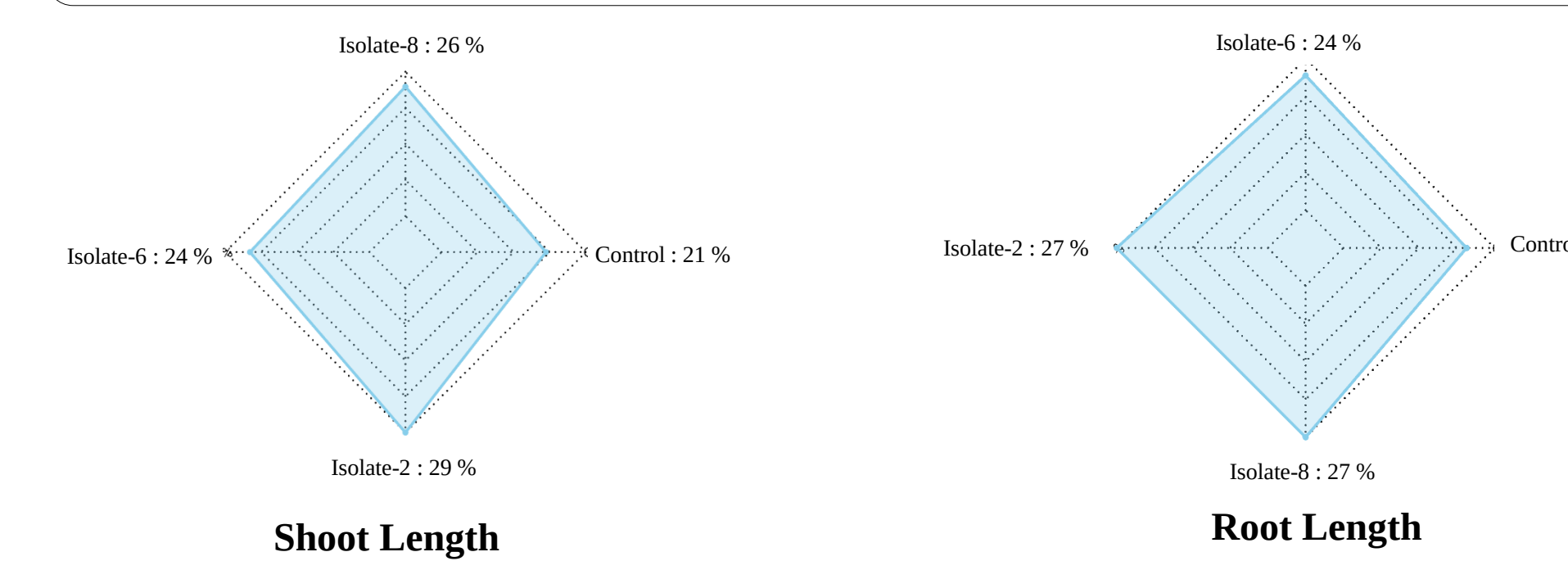
Graphical Representation of Survival Rate (%) of Isolated Bacteria in Presence of Mn (mg/L)



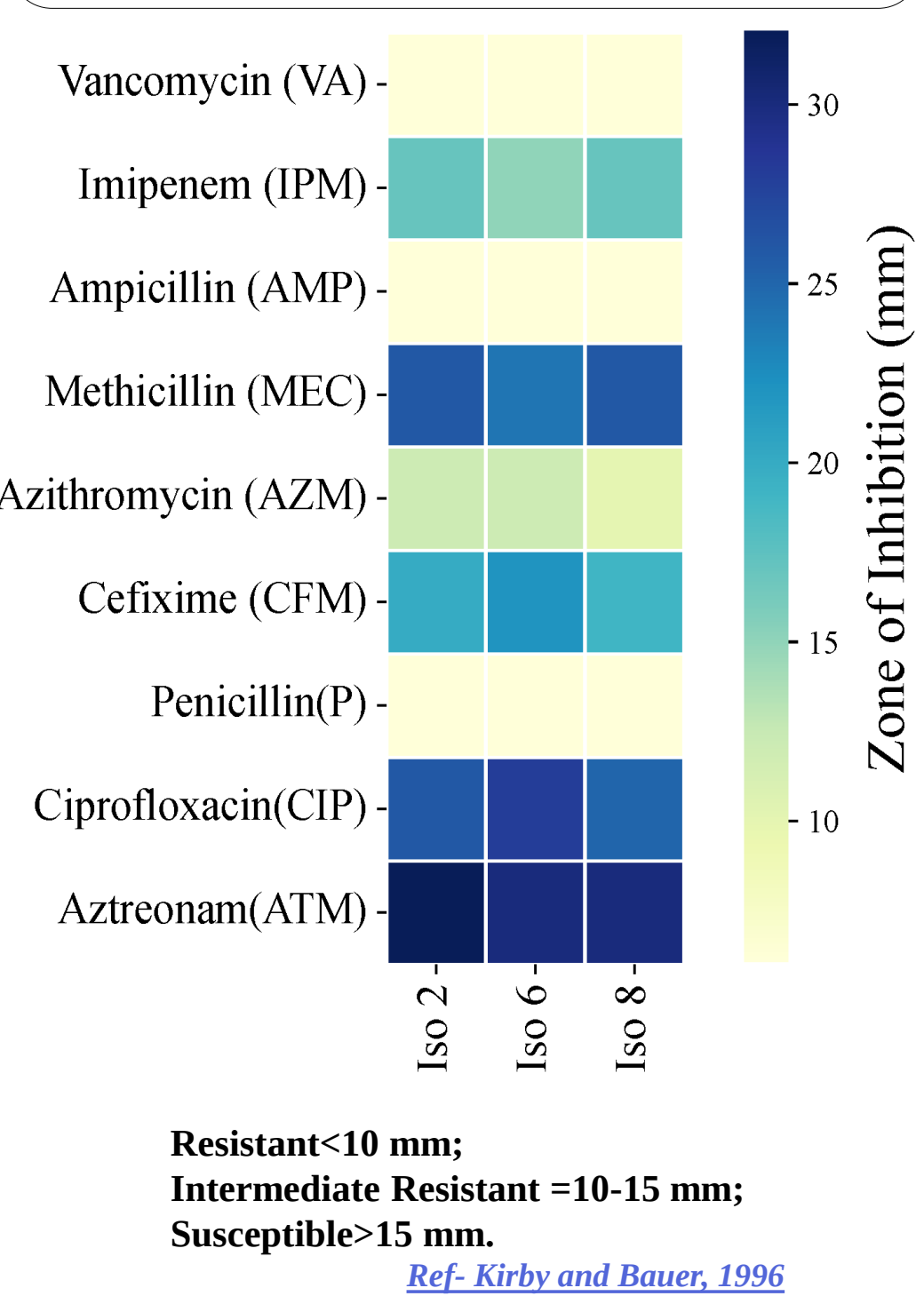
Graphical representation of Heavy Metal Removal Rate (%) of Isolated Bacteria Using AAS



Graphical Representation of The Effect of Isolated Bacteria on Seedling Growth



Graphical Representation of Antibiotic Sensitivity Test Results



CONCLUSION

This study successfully isolated and characterized heavy metal-resistant plant growth-promoting rhizobacteria (PGPR) from the rhizosphere of vegetable plants in Savar, Dhaka, an area significantly affected by industrial contamination. All of the nine bacterial isolates were investigated for their heavy metal resistance and plant growth-promoting (PGP) characteristics. Strong PGP activities, including nitrogen fixation, phosphate and zinc solubilization, ammonia generation, indole-3-acetic acid (IAA) synthesis, and siderophore secretion, were demonstrated by three isolates: Iso-2, Iso-6, and Iso-8. These isolates also showed exceptional resistance to lead (Pb), manganese (Mn), and cadmium (Cd); Iso-6 and Iso-8 had the greatest metal removal capabilities (63%–86%). Pot experiments with rice (BRRI dhan28) further validated the effectiveness of these isolates in promoting plant growth under heavy metal stress. Bacterial inoculation significantly enhanced shoot and root length, biomass accumulation, and overall seedling vigor compared to non-inoculated controls. The ability of these rhizobacteria to improve plant health despite the presence of toxic metals highlights their potential as natural biofertilizers and bioremediators.

REFERENCES

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- Kumar RS, Koner S, Tsai H-C, Chen J-S, Huang S-W, Hsu B-M. Deciphering endemic rhizosphere microbiome community's structure towards the host-derived heavy metals tolerance and plant growth promotion functions in serpentine geo-ecosystem. *Journal of Hazardous Materials*. 2023;452:131359.
- Khalil MMR, Fierro-Coronado RA, Peñuelas-Rubio O, Villa-Lerma AG, Plascencia-Jatomea R, Félix-Gastélum R, et al. Rhizospheric bacteria as potential biocontrol agents against Fusarium wilt and crown and root rot diseases in tomato. *Saudi Journal of Biological Sciences*. 2021;28(12):7460-71.